

Northeastern University

Principles and Applications of

Remote Sensing (Bilingual)

Computer Practice Report



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April 21, 2025

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1. Computer experiment content (1): band conversion, radiation correction

1. Experimental content

① Data preparation: Download Landsat or Sentinel images with an interval of more than five years between two scenes, and the image area is your hometown or other areas of interest. Requirements: At least one of the downloaded data is new data in recent years; Clear and good-quality images without clouds are needed; You need to try to download images in the same month or quarter; It needs to include 6 types of land use types: cultivated land, forest land, grassland, water body, construction land and unused land. If there are less than 6 types of land features in the study area, please select another area!

② Radiation calibration: Find the calibration parameters, select the radiation calibration tool, set the calibration parameters, and execute the radiation calibration

③ Atmospheric correction: Open the FLAASH module, input images and parameters, set output parameters, perform atmospheric correction, remove the influence of the atmosphere, and calculate the true reflectivity of the surface.

④ View the correction results: select at least three ground features and describe the differences in pixel values of the three ground features in different wavelength bands.

2. Experimental purpose

① Understand the band characteristics of remote sensing images: master the band composition and physical significance of multispectral remote sensing images, and understand the differences in the response of different bands to ground features.

② Master the principles and methods of radiation correction: learn the principles of radiation calibration and atmospheric correction, and master the technical process of eliminating sensor errors and atmospheric interference.

③ Proficient in remote sensing software operation skills: Complete band combination transformation and radiation correction operations through tools such as ENVI or ArcGIS to improve remote sensing data processing capabilities.

④ Cultivate quantitative analysis ability: analyze the influence of radiation correction on remote sensing quantitative inversion (such as vegetation index and surface temperature) by comparing images before and after correction.

3. Experimental principle

Radiation correction is to eliminate or reduce the radiation errors caused by sensor response characteristics, atmospheric scattering and absorption, illumination conditions and other factors in remote sensing images, so that the images can truly reflect the reflection or emission characteristics of ground objects.

The response function of the sensor may be non-linear and need to be corrected. Let the input radiance of the sensor be L and the output digital quantization value be DN . Generally, the relationship between the two is determined by laboratory calibration or site calibration, usually expressed by a polynomial function:

$$DN = a_0 + a_1L + a_2L^2 + \cdots + a_nL^n$$

Acquisition of coefficients by calibration $a_0, a_1, \cdots, a_n$ Thereby realizing the correction of the sensor response. Atmospheric correction. The scattering and absorption of electromagnetic waves by the atmosphere will cause radiation errors in

remote sensing images.

Commonly used atmospheric correction methods include the method based on radiative transfer model and empirical linear method. In the method based on radiation transfer model, by considering the scattering, absorption and reflection of the atmosphere, the radiation transfer equation is established to calculate the true reflectance of ground objects. The simplified radiative transfer equation is

$$L = \frac{\pi L_{path} + \rho E_{solar} T}{\pi - \rho S}$$

where L is the radiance received by the sensor, L_{path} is atmospheric radiance, E_{solar} is the incident solar irradiance, T is the atmospheric transmittance, and S is the atmospheric scattering coefficient.

By measuring or estimating the parameters in the equation, the true reflectance of the ground object can be inverted, thereby achieving atmospheric correction.

4. Experimental results and analysis

4.1. Analysis of atmospheric correction results of 2014 data:



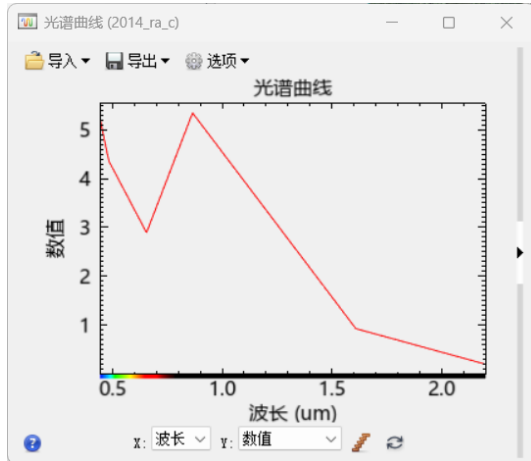
(a) Remote sensing images
before atmospheric correction in 2014



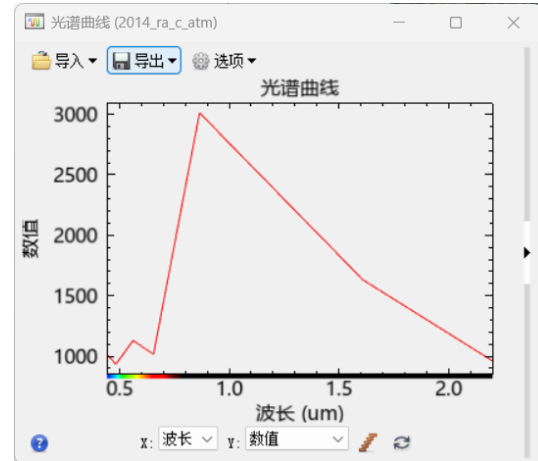
(b) Atmospheric corrected
remote sensing images in 2014

The overall clarity of the image after atmospheric correction is improved compared with that before correction. The image before correction has a certain degree of haziness, as if it is covered by a layer of tulle, which is caused by atmospheric scattering and other factors. After correction, this hazy feeling is weakened, the outline of ground objects is clearer, and the details are easier to distinguish. Before correction, due to the influence of the atmosphere, the color may be deviated, and the color in some areas appears too dim or distorted. After correction, the color of ground features is more in line with the actual situation, and the tone difference between different ground features is more obvious, which is convenient to distinguish different ground feature types.

4.1. 1. Forest change:



(a) Forest band curves before
atmospheric correction in 2014

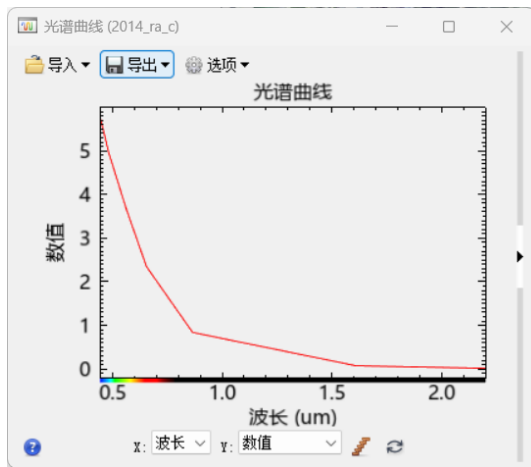


(b) Forest band curves after
atmospheric correction in 2014

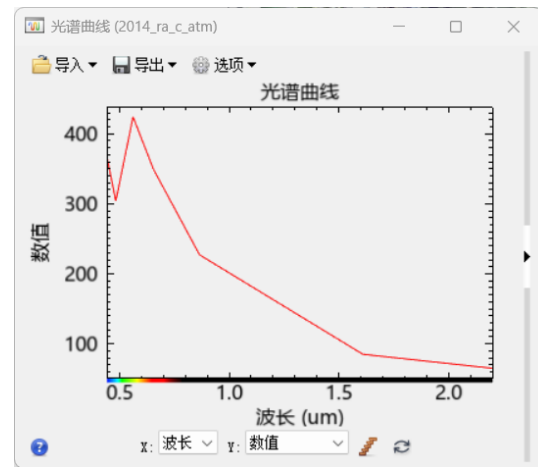
Compared with the forest spectral curves before and after atmospheric correction, the difference is significant. Numerically, the magnitude before correction is small, but it increases greatly after correction, which is closer to the actual radiation level. In terms of shape, the trough in the short-wave region is

abrupt before correction, which is greatly affected by atmospheric scattering and absorption. After correction, the trough is gentle, and the true reflection can be revealed; The peak position is approximate, and the height increases sharply after correction, highlighting the high reflectivity of the forest in the corresponding band; In the long-wave region, the decline slows down before correction, but the change is obvious after correction, which accurately shows the characteristics of decreasing reflectivity of forest in this band. Atmospheric correction effectively eliminates atmospheric interference, so that the forest spectral curve can accurately reflect its reflection characteristics.

4.1. 2. Water body changes:



(a) Water band curves before
atmospheric correction in 2014

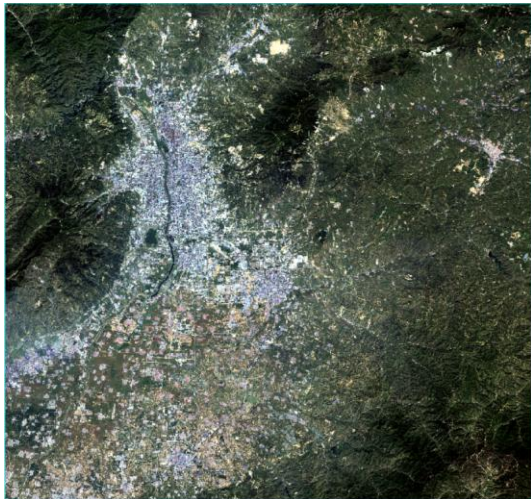


(b) Water band curves after
atmospheric correction in 2014

Comparing the spectral curves of water body before and after atmospheric correction, in terms of numerical magnitude, the curve values before correction are concentrated at 0-5, and rise to about 0-400 after correction, which is more in line with the actual radiation intensity of water body after correction. In the shape of the curve, although there is a peak value in the short-wave region before correction, the

value is low, but after correction, the peak value is obvious and the value is greatly improved; In the long-wave region, the decline is slow before correction, and the decline trend is steeper after correction. Atmospheric correction eliminates atmospheric interference, allows the spectral curve of water body to more accurately reflect its reflection characteristics in different wavelength bands, and improves the ability of spectral data to characterize the true radiation characteristics of water body.

4.2. Analysis of atmospheric correction results of 2020 data:



(a) Remote sensing images
before atmospheric correction in 2020

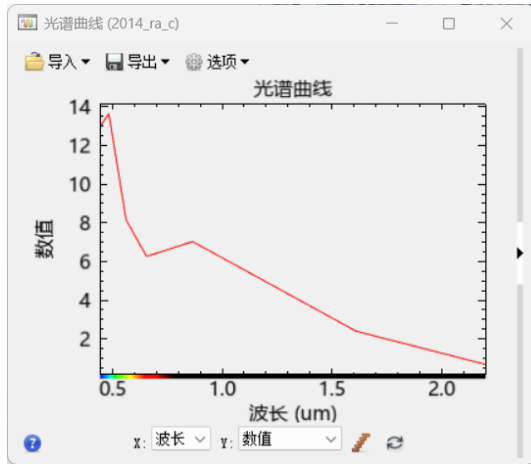


(b) Remote sensing imagery
after atmospheric correction in 2020

Before correction, the image is affected by atmospheric scattering and absorption, the reflectivity representation of ground objects is inaccurate, the color and clarity are poor, and the details are disturbed. After correction, the influence of the atmosphere is weakened, the spectral information of ground objects is more realistic, the image color and contrast are more reasonable, the boundaries and

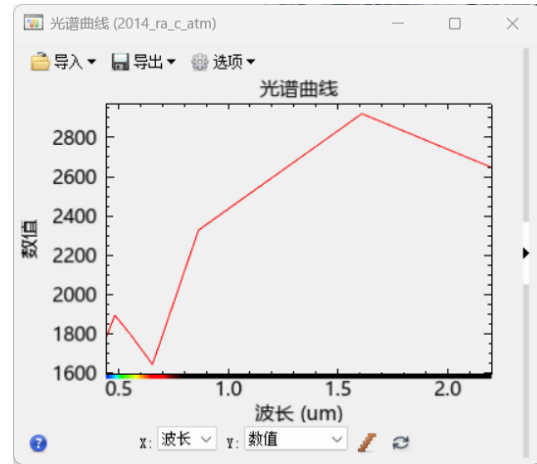
details of ground objects are clearer and easier to distinguish, and the image quality is improved.

4.2. 1. Architectural changes:



(a) Building band curves

before atmospheric correction in 2014



(b) Building band curves

after atmospheric correction in 2014

Compared with the architectural spectral curves before and after atmospheric correction, the difference is obvious. Numerically, before correction, the curve value is concentrated at 0-14, and after correction, it jumps to about 1600-2800, which can better reflect the actual radiant energy of the building. In terms of shape, there is an obvious trough in the short-wave region before correction, and after correction, the trough becomes shallower and the value increases greatly; In the long-wave region, the decline is slow before correction, and the decline trend is more significant after reaching the peak after correction. Atmospheric correction effectively removes the interference of atmospheric radiation to buildings, makes the spectral curve more accurately reflect the reflection characteristics of buildings in different wavelength bands, and enhances the ability of spectral data to present the real radiation situation of buildings.

2. Contents of computer experiments (2): Supervised classification of remote sensing images

1. Experimental content

① Data preparation: high-quality remote sensing images after atmospheric correction, cropped images after radiation correction, the cropped image features should be as rich as possible, and the cropped size should not be less than 50km * 50km. Use multiple (at least three) supervised classification methods for classification (cultivated land, woodland, grassland, water body, building land, unused land), compare the classification accuracy, select the best classification method and map it.

② Sample selection: According to the ground truth data on the remote sensing image, the representative ground object samples are selected as training samples. The samples should cover all types of features of interest in the study area, and have sufficient quantity and distribution uniformity to ensure that the classifier can accurately learn the spectral characteristics of different features.

③ Supervised classification selection: According to the characteristics and research purpose of remote sensing images, select appropriate supervised classifiers, such as maximum likelihood classifier, random forest, etc.

④ Unsupervised classification selection: ISODATA classification method was adopted.

⑤ Accuracy evaluation: Use the reserved ground truth data or independent verification samples to evaluate the accuracy of classification results. Commonly

used accuracy evaluation indexes include overall accuracy, classification accuracy, Kappa coefficient, etc.

⑥ Mapping. Map mapping with ArcGIS.

2. Experimental purpose

① Understand the basic principles of supervised classification: master the core concepts of supervised classification (such as training samples, classifiers, feature spaces) and understand the differences between supervised classification and unsupervised classification.

② Master the technical process of supervised classification: learn the complete process from sample selection, classifier training to result post-processing, and be familiar with classification algorithms (such as maximum likelihood, random forest, etc.).

③ Improve the interpretation ability of remote sensing images: Train ground feature recognition and classification skills through practical cases (such as land use/cover classification) to enhance the understanding of spectral features and spatial features.

④ Master the evaluation method of classification accuracy: Learning the evaluation indexes such as confusion matrix, overall accuracy and Kappa coefficient can quantitatively analyze the reliability of classification results.

3. Experimental principle

3.1. Maximum likelihood method

Maximum likelihood method is a common supervised classification method of

remote sensing images, which is based on probability theory and Bayesian decision theory. The basic hypothesis assumes that the spectral features of objects in the training area approximately obey a normal distribution. Based on this, the mean vector of each category can be calculated $\bar{X}_i$ covariance matrix, C_i And other statistical parameters, and then deduce the prior probability density function of the population.

For the pixel X to be classified, the conditional probability of belonging to each class is calculated according to the Bayesian formula $P(i|X)$ I'm Taking multi-band remote sensing images as an example, pixel X is a vector containing multi-band spectral values. Assuming that there are n classes, it is calculated by the probability density function of each class $P(X|i)$ (probability of occurrence of pixel X under the i-th condition), combined with a prior probability $P(i)$ (probability of occurrence of the i-th class in the population), according to the Bayesian formula

$$P(i|X) = \frac{P(X|i)P(i)}{\sum_{j=1}^n P(X|j)P(j)}$$

The probability that the pixel X belongs to the i-th class is calculated. Partition of pixels to attribution probabilities $P(i|X)$ Among the largest categories. If there is a large difference between the overall distribution and the normal distribution, the classification accuracy of this method will be affected.

3.2. ISODATA

The initial setting determines the initial clustering center and a series of initial parameters, such as the desired number of clusters k, the initial number of clusters k, and the minimum number of samples per category threshold θN , sample feature maximum standard deviation threshold θs Minimum distance threshold between

category centers θ_c , the maximum merging logarithm L in a merging operation, and the number of iteration operations I .

The minimum distance method is used to divide the samples into the cluster block where the nearest cluster center is located. The number of samples removed is less than θ_{NC} Category of. Update the cluster center, and recalculate the sample mean in each cluster as the new center. If the standard deviation of a certain sample characteristic is greater than θ_s , indicating that there are large differences within the class, and the class is split into two categories. If that distance between the center of the two categories is less than θ_c , indicate that the two categories are similar, and merge them.

Iteration termination continuously repeats the above-mentioned clustering and category adjustment steps until the maximum iteration number I is reached, or the clustering result meets the set classification parameters (such as the category no longer changes, etc.), and classification is completed.

3.3. Random Forest

Decision tree-based learner. According to the spectral characteristics and other attributes of remote sensing image pixels, the decision tree divides the data set step by step through a series of node splitting rules, and each leaf node corresponds to a category judgment. For example, when distinguishing vegetation from buildings, node splitting decisions are made based on characteristics such as reflectivity in the near infrared band.

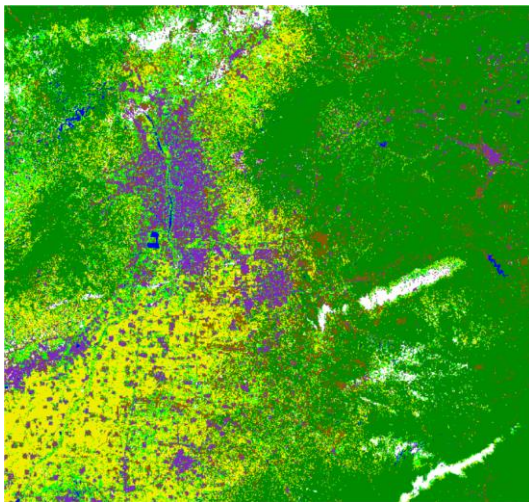
Bootstrap sampling is used to randomly select multiple subsets similar in size

to the original training set from the original training set, and each subset is used to construct a decision tree. In this way, different decision tree training sets are different, which makes the model more diverse. When constructing the node splitting of each decision tree, instead of considering all features, a subset is randomly selected from all features, and then the optimal feature is selected from this subset for splitting. This reduces the correlation between trees and enhances the model generalization ability.

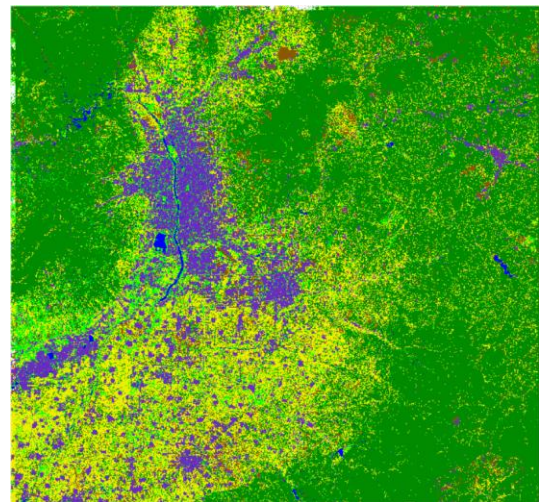
After the construction of multiple decision trees, when classifying unknown pixels, each decision tree gives the classification results, and the final category of pixels is determined by majority voting method (classification task). That is, all decision tree classification results are counted, and the category with the most votes is the category to which the pixel belongs.

4. Experimental results and analysis

4.1. Maximum likelihood method



(a) Maximum likelihood classification in 2014



(b) Maximum likelihood classification in 2020

Class Confusion Matrix

文件

Confusion Matrix: D:\Person\学习\遥感\上机\output\2014\2014_as

Overall Accuracy = (24379/27159) 89.7640%

Kappa Coefficient = 0.8229

Class	Ground Truth (Pixels)			
Unclassified	cloud	glass	water	forest
cloud	2299	3	0	11
glass	0	302	45	556
water	0	14	671	63
forest	0	5	0	16287
unuse land	8	3	0	240
fara	0	158	0	180
urban	39	0	25	32
Total	2346	492	741	17365

Class	Ground Truth (Pixels)				Total
Unclassified	fara	urban			
cloud	0	0			2350
glass	277	184			1364
water	0	18			766
forest	452	0			16744
unuse land	119	273			1458
fara	2380	0			2718
urban	3	1625			1759
Total	3242	2109			27159

Class	Ground Truth (Percent)			
Unclassified	cloud	glass	water	forest
cloud	0.00	0.00	0.00	0.00
glass	98.00	0.61	0.00	0.06
water	0.00	61.38	6.07	3.20
forest	0.00	2.85	98.55	0.36
unuse land	0.00	1.02	0.00	93.79
fara	0.00	0.34	0.61	1.38
urban	1.66	32.11	0.00	1.04
Total	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)				Total
Unclassified	fara	urban			
cloud	0.00	0.00			0.00
glass	98.00	0.61			0.06
water	0.00	61.38			6.07
forest	0.00	2.85			0.36
unuse land	0.00	1.02			93.79
fara	0.00	0.34			1.38
urban	1.66	32.11			1.04
Total	100.00	100.00			100.00

(c) Maximum likelihood accuracy and kappa in 2014

Class Confusion Matrix

文件

Confusion Matrix: D:\Person\学习\遥感\上机\output\2020\2020_as

Overall Accuracy = (8444/10358) 81.5215%

Kappa Coefficient = 0.7675

Class	Ground Truth (Pixels)			
Unclassified	glass	forest	urban	fara
glass	274	1	294	134
forest	7	1468	4	898
urban	7	0	2774	53
fara	74	27	47	1879
unuser land	28	0	35	0
Total	392	1496	3236	3036

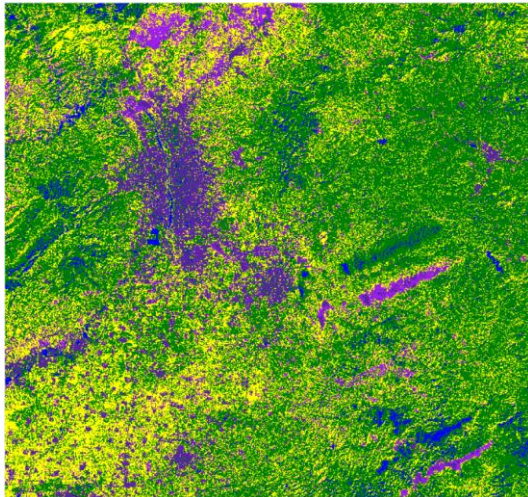
Class	Ground Truth (Pixels)				Total
Unclassified	unuse land				
glass	12				766
forest	1				2378
urban	19				2860
fara	59				2086
unuser land	0				941
Total	1171				13277

Class	Ground Truth (Percent)			
Unclassified	glass	forest	urban	fara
glass	0.00	0.00	0.00	0.00
forest	69.90	0.07	9.09	4.41
urban	1.79	98.13	0.12	29.58
fara	1.79	0.00	85.72	1.75
unuser land	18.88	1.80	1.45	61.89
unuser land	7.14	0.00	3.08	0.00
Total	100.00	100.00	100.00	100.00

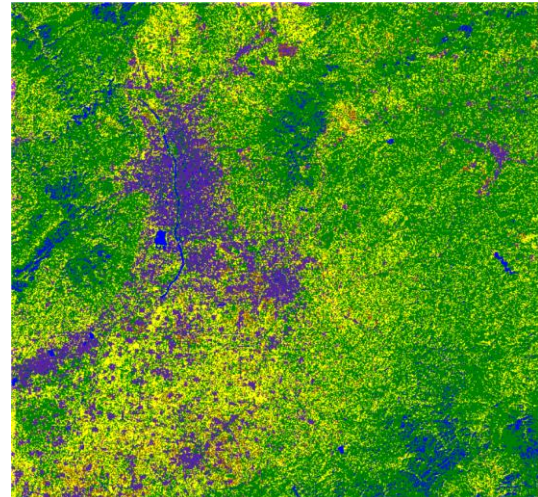
Class	Ground Truth (Percent)				Total
Unclassified	unuse land				
glass	0.00				0.00
forest	0.95				7.40
urban	0.08				22.96
unuser land	1.51				27.61
Total	100.00				100.00

(d) Maximum likelihood accuracy and kappa in 2020

4.2. ISODATA



(a) 2014 ISODATA classification



(b) 2020 ISODATA classification

Class Confusion Matrix

文件

Confusion Matrix: D:\Person\学习\遥感\上机\output\2014\2014_isodata_10_class

Overall Accuracy = (17408/23949) 72.6878%

Kappa Coefficient = 0.4632

Class	Ground Truth (Pixels)			
Unclassified	water	fara	urban	glass
water	672	0	31	105
fara	0	1740	35	131
urban	24	3	1802	2
glass	0	0	73	9
forest	45	0	168	251
Total	741	3242	2109	492

Class	Ground Truth (Pixels)				Total
Unclassified	water	fara	urban	glass	
water	2728				4023
fara	4023				1840
urban	1840				232
glass	15126				15126
forest	23949				23949

Class	Ground Truth (Percent)			
Unclassified	water	fara	urban	glass
water	0.00	0.00	0.00	0.00
fara	90.69	0.00	1.47	21.34
urban	0.00	53.67	1.66	26.63
glass	3.24	0.09	85.44	0.41
forest	0.00	0.06	3.46	0.61
Total	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)				Total
Unclassified	water	fara	urban	glass	
water	11.39				16.80
fara	16.80				7.68
urban	7.68				0.97
glass	0.97				63.16
forest	63.16				63.16

(c) 2014 ISODATA accuracy and kappa

Class Confusion Matrix

文件

Confusion Matrix: D:\Person\学习\遥感\上机\output\2020\2020_isodata_10_class

Overall Accuracy = (18504/25242) 73.3064%

Kappa Coefficient = 0.5074

Class	Ground Truth (Pixels)			
Unclassified	fara	urban	forest	unuse land
fara	1489	85	2004	145
city #	62	2672	29	535
forest #	1340	366	13588	74
unuse land #	144	15	79	95
water	1	98	1665	15
Total	3036	3236	17365	864

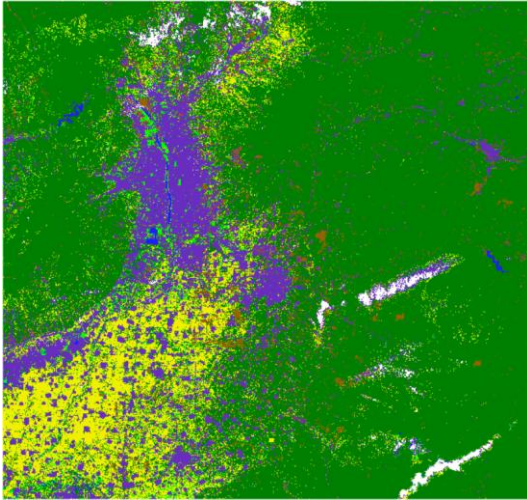
Class	Ground Truth (Pixels)				Total
Unclassified	fara	urban	forest	unuse land	
fara	3724				3327
city #	15419				15419
forest #	393				2439
unuse land #	0				25242

Class	Ground Truth (Percent)			
Unclassified	fara	urban	forest	unuse land
fara	0.00	0.00	0.00	0.00
city #	49.04	2.53	11.54	16.78
forest #	2.04	82.57	0.17	61.92
unuse land #	44.14	11.31	78.25	9.56
unuse land #	4.74	0.46	0.45	11.00
water	0.03	3.03	9.59	1.74
Total	100.00	100.00	100.00	100.00

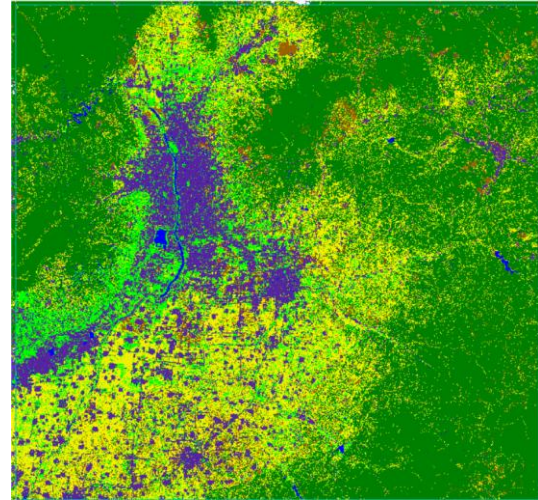
Class	Ground Truth (Percent)				Total
Unclassified	fara	urban	forest	unuse land	
fara	0.00				14.75
city #	13.18				61.08
forest #	1.32				1.32
unuse land #	9.66				9.66
Total	100.00				100.00

(d) 2020 ISODATA accuracy and kappa

4.3. Random Forest



(a) 2014 Random Forest Classification



(b) 2020 Random Forest Classification

Class Confusion Matrix

文件

Confusion Matrix: D:\Person\学习\遥感\上机\output\2014\for

Overall Accuracy = (25072/27159) 92.3156%

Kappa Coefficient = 0.8577

Class	Ground Truth (Pixels)	fara	unuse land	forest	water
Unclassified	0	0	0	0	0
urban	2042	54	125	33	24
fara	1709	0	56	0	0
unuse land	25	23	736	48	0
forest	7	1338	3	17148	0
water	8	0	0	7	716
glass	17	118	0	73	1
cloud	9	0	0	0	0
Total	2109	3242	864	17365	741

Class	Ground Truth (Pixels)	cloud	Total
Unclassified	0	0	0
urban	18	35	2331
fara	29	0	1795
unuse land	0	9	841
forest	10	8	18514
water	6	0	737
glass	428	1	638
cloud	1	2293	2303
Total	492	2346	27159

Class	Ground Truth (Percent)	fara	unuse land	forest	water
Unclassified	0.00	0.00	0.00	0.00	0.00
urban	96.82	1.67	14.47	0.19	3.24
fara	0.05	52.71	0.00	0.32	0.00
unuse land	1.19	0.71	85.19	0.28	0.00
forest	0.33	41.27	0.35	98.75	0.00
water	0.38	0.00	0.00	0.04	96.63
glass	0.81	3.64	0.00	0.42	0.13
cloud	0.43	0.00	0.00	0.00	0.00
Total	100.00	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)	glass	cloud	Total
Unclassified	0.00	0.00	0.00	0.00
urban	96.82	1.67	14.47	98.56
fara	0.05	52.71	0.00	52.76
unuse land	1.19	0.71	85.19	87.09
forest	0.33	41.27	0.35	41.95
water	0.38	0.00	0.00	0.38
glass	0.81	3.64	0.00	4.45
cloud	0.43	0.00	0.00	0.43
Total	100.00	100.00	100.00	100.00

(c) 2014 Random Forest accuracy and kappa

Class Confusion Matrix

文件

Confusion Matrix: D:\Person\学习\遥感\上机\output\2020\forest

Overall Accuracy = (9111/10358) 87.9610%

Kappa Coefficient = 0.8467

Class	Ground Truth (Pixels)				
	water	fara	urban	forest	glass
Unclassified	0	0	0	0	0
water	895	4	30	0	24
fara	0	2337	13	124	33
urban	5	82	2997	1	12
forest	0	173	4	1365	0
glass	36	328	130	6	323
unuser_land	0	102	62	0	0
Total	936	3036	3236	1496	392

Class	Ground Truth (Pixels)	
	unuse land	Total
Unclassified	0	0
water	0	953
fara	37	2544
urban	18	3115
forest	0	1542
glass	13	846
unuser_land	1194	1358
Total	1262	10358

Class	Ground Truth (Percent)				
	water	fara	urban	forest	glass
Unclassified	0.00	0.00	0.00	0.00	0.00
water	95.62	0.13	0.93	0.00	6.12
fara	0.00	76.98	0.40	8.29	8.42
urban	0.53	2.70	92.61	0.07	3.06
forest	0.00	5.70	0.12	91.24	0.00
glass	3.85	11.13	4.02	0.40	82.40
unuser_land	0.00	3.36	1.92	0.00	0.00
Total	100.00	100.00	100.00	100.00	100.00

Class	Ground Truth (Percent)	
	unuse land	Total
Unclassified	0.00	0.00
water	0.00	9.20
fara	2.93	24.56
urban	1.43	38.07

(d) 2020 Random Forest accuracy and kappa

4.4. Change Analysis

Judging from these sets of classification results images, the ground features changed significantly from 2014 to 2020.

Urban area expansion: The extent of urban areas represented by purple or dark colors is expanded in the maximum likelihood method, ISODATA, and random forest classification images. The change of pixel category shows that the city is expanding outward, which may be the construction of new buildings, roads and other

infrastructure, reflecting the acceleration of urbanization.

Vegetation cover changes: green areas increase and decrease. The decrease of vegetation in some areas may be due to the encroachment of urban construction and agricultural development; There are also areas where the green scope has increased, which may be the effect of afforestation and ecological restoration projects.

Land use type conversion: The change of pixel distribution of different colors indicates the land use type conversion. For example, agricultural land (yellow area), urban land and vegetation land are converted.

4.5. Comparison of classification methods

Maximum likelihood method: From the image, in terms of land use type distinction, the classification boundaries of land features with relatively obvious spectral characteristics such as cities and water bodies are relatively clear. For example, in the images of 2014 and 2020, the water body part is relatively regular. This is because it is based on probability statistics and can use spectral characteristic probability distribution to effectively distinguish different feature categories on data that conforms to the normal distribution assumption.

ISODATA: It can be found in the image that for some complex areas with ambiguous land use types, the clustering can be automatically adjusted. For example, in the urban-suburban junction, the category division in images was constantly adjusted from 2014 to 2020, reflecting its ability to automatically adapt to data distribution and discover potential category structures.

Disadvantages: From the overall image, the category division is sometimes not fine

3. Content of computer experiment (3): Analysis of land use change

1. Experimental content

① Data preparation and land use classification: Select the best classification method of the last experimental class to classify another image.

② Calculation of land use change: extract the information of land use change and calculate the area.

③ Analysis of land use change: Analyze the transformation of land use types in the two images in the study area, which land type changes to which land type the most, and what are the main driving factors.

2. Experimental purpose

① Master the core methods of land use change detection: understand the principle of multi-temporal remote sensing image change detection, and be familiar with pixel-based and object-oriented change analysis methods.

② Improve multi-temporal data processing capabilities: learn the preprocessing (radiation correction, geometric registration) and data consistency control of two or more remote sensing images.

③ Cultivate spatio-temporal analysis skills: master the technical process of land use dynamic monitoring through change patch extraction, change type classification and driving force analysis.

④ Strengthen practical application ability: Combined with cases such as urban expansion and deforestation, understand the application value of remote sensing in resource management and ecological assessment.

3. Experimental principle

First of all, it is necessary to collect land use data in different periods, which can be vector data obtained by interpreting remote sensing images or statistical data from land survey. Then the data is preprocessed, including the unification of coordinate system, the conversion of data format and the cleaning of data, so as to remove wrong or invalid records, ensure the accuracy and consistency of data, and lay the foundation for subsequent analysis.

The land use vector data of different periods are superimposed. For example, use the Union tool to merge two epochs of land use layers to produce a new layer that contains information about all polygons and their attributes. In this new layer, by comparing the attributes of land use types of each polygon in different periods, we can intuitively find the changes of land use types. For example, if the attribute of a plot is "cultivated land" in the early stage, and it becomes "construction land" in the later stage, it can be determined that the land use type of the plot has changed.

Using the summary statistical tool in ArcMap, the area of different land use types is counted. According to the classification fields of land use types, the total area of each type in different periods can be calculated respectively. By comparing the area data of various land use types in different periods, we can understand the scale change trend of various land use types, such as whether the cultivated land area increases or decreases, the expansion speed of construction land area, etc.

Through a series of data processing operations, combined with the pivot table function of Excel, the land use transfer matrix is constructed. Firstly, the land use

data of different periods are intersected, and the cross information of land use types of each plot in different periods is obtained. Then, the fusion operation is used to merge the plots with the same land use type conversion, and their areas are counted. Finally, these data are collated into Excel pivot table to form a land use transfer matrix. This matrix can clearly show the destination of the decrease and the source of the increase of various land use types. For example, it can be clearly seen how much cultivated land is converted into forest land, and how much of the increase of forest land comes from the conversion of other land use types.

4. Experimental results and analysis

4.1. Matrix results

2014年 \ 2020年	草地	裸土	林地	建筑	耕地	水体	2020年总计
草地	22.8248997	18.9071999	142.5500946	18.7730999	55.3707008	0.4644	258.8903949
裸土	161.0747986	30.9204006	251.5823975	119.0330963	249.1172943	9.8541002	821.5820875
林地	4.5783	0.4329	16.3017006	4.6691999	2.2635	24.0002995	52.2459
建筑	82.2410965	90.7406998	3236.321777	34.7391014	467.2898865	5.9714999	3917.304061
耕地	86.129097	75.8780975	122.1435013	236.9187012	231.7823944	3.4335001	756.2852915
水体	140.8482056	70.8128967	396.9432068	61.0713005	643.6242065	2.2284	1315.528216
2014年总计	497.6963974	287.6921945	4165.842678	475.2044992	1649.447983	45.9521997	7121.835951

4.2. Matrix analysis

From the transfer matrix data, it can be seen that the land use pattern has changed significantly in these six years.

Ecological changes: The area of forest land has increased significantly, and a large number of bare soil and grassland have been transferred to forest land, indicating that afforestation, ecological restoration and other projects have been fruitful, and the ecological environment has been improved to a certain extent. However, the reduction of water area and conversion into cultivated land,

buildings, etc. may affect the regional ecological balance and hydrological cycle, so it is necessary to pay attention to the protection and rational utilization of water resources.

Urbanization process: Construction land has expanded rapidly, and a large number of land have been transferred from forest land, bare soil, cultivated land, etc., which shows that urban construction is fast and the demand for land is strong. This is the result of economic development and population agglomeration, but it also brings problems such as the reduction of cultivated land and the compression of ecological space. It is necessary to balance construction and ecological protection in urban planning.

Agricultural development trends: The area of cultivated land fluctuates, and some grasslands and bare soil are converted into cultivated land, reflecting agricultural development activities. At the same time, cultivated land has also been converted to construction and other land, reflecting that agricultural land is squeezed by urbanization. It is necessary to coordinate the relationship between agriculture and construction land to ensure food security and sustainable agricultural development.